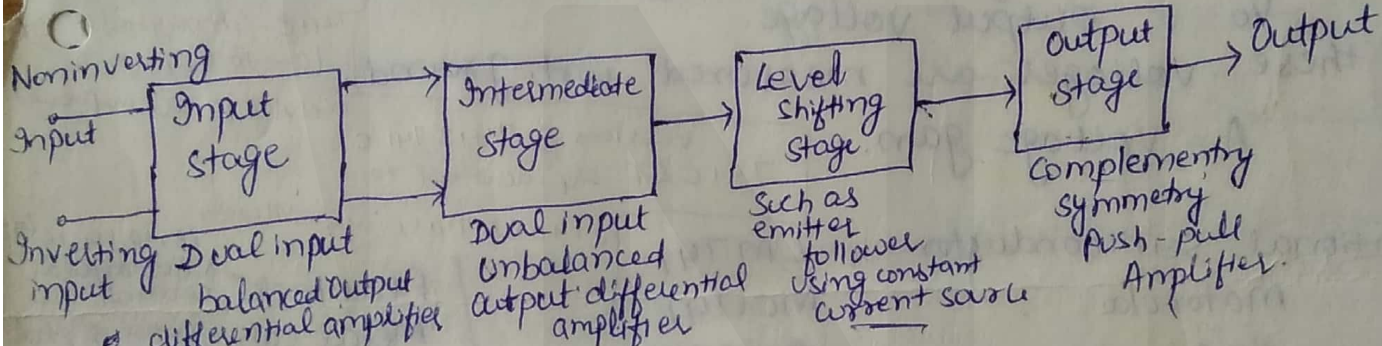
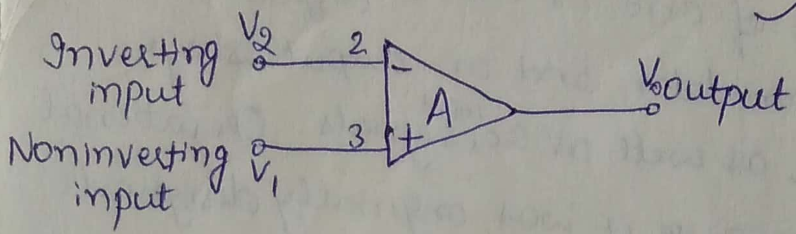
Operational Amplifier :- is a direct coupled, high gain amplifier usually consisting of one or more differential amplifier and followed by a level translator and an output stage. It can be used to amplify dc as well as ac input signals. Operational amplifier is named such as because it was originally designed for mathematical operation such as addition, subtraction, integration and differentiation. Now a days it can be used for dc and ac signal amplification, active filters, oscillators, comparators, regulators etc.

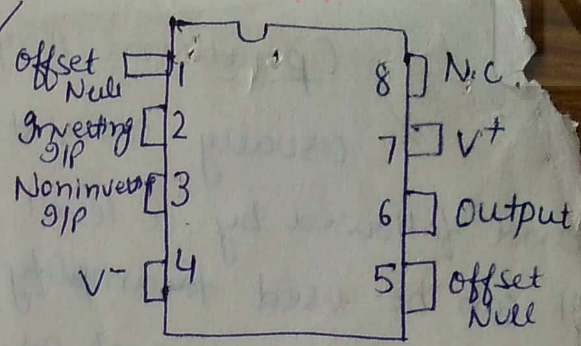


Block Diagram of an op-amp.

The first stage provide most of the voltage (60dB) gain of the amplifier. & it provides the ~~most of voltage~~ input resistance of op amp. The intermediate stage is usually driven by output of first stage. Next stage is level shifting stage (2 gain). As direct coupling is used, the dc voltage at output of intermediate stage is well above the ground potential. Hence a level shift stage is used after the intermediate stage to shift the dc level at O/P of intermediate stage ~~to shift the dc level at output of inter~~ downward to zero volts w.r.t. ground. The output stage is usually a complementory amplifier output stage. The output stage increases the output voltage swing & raising the current supplying capability of op amp. Output of opamp It also establishes the low output resistance. is measured at output of this final stage push pull amplifier w.r.t. ground.

Symbol of Op-amp :-





The ac signal (dc voltage) applied to ~~this~~ (+) input produces an in-phase (same polarity) signal at output. On the ~~othe~~ other hand (−) input produces an 180° out of phase (opposite polarity) signal at o/p.

V_1 = voltage at noninverting input
V_2 = voltage at inverting input
V_o = output voltage.

All these voltages are measured w.r.t. ground.

A = Voltage gain.

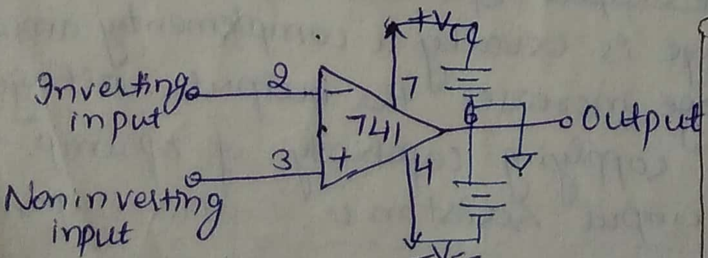
A, C, E, S & SC
741 - Millitary grade op-amp (−55° to 125°C)
741C - Comm. grade op-amp (0 to 70/75°C)
741 A & E are improved versions of 741 & 741C
741C & E are identical to
741 S & 741SC are millitary

National semiconductor = LM741
Motorola MC1741
RCA CA3741
Texas instruments SN52741
Signetics N5741

Flat Pack - rectangular ceramic case (8, 10, 14, 16 pins)
Metal can - metal & plastic case 3, 5, 8, 10, 12
Dual in line Package (DIP) mounted inside a plastic or ceramic case. It is mostly used ∵ it can be mounted easily. 8 pin DIP. 12, 14, 16, 20.

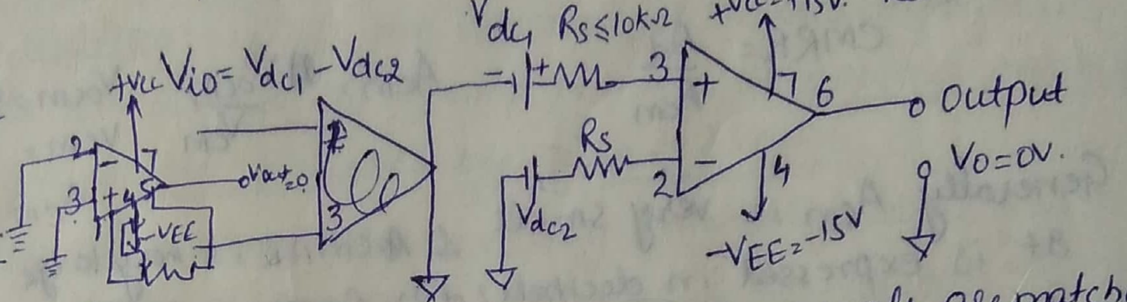
Power Supplies Connections :-

For operation of op amp bipolar voltages are required i.e. two voltages are required, one +ve & other equal in magnitude but negative. These voltages must be referenced to a common point or ground. Fairchild :- ($V+$, $V-$), Motorola uses. ($+V_{CC}$ & $-V_{EE}$).



Temp. Range
Millitary temp. −55 to 125°C
Industrial −20 to 85°C
Comm. temp. Range − 0 to 70°C

DC char. (1) **Input offset Voltage** (V_{io}) :- It is the voltage that must be applied between two input terminal of an op amp to null the output.

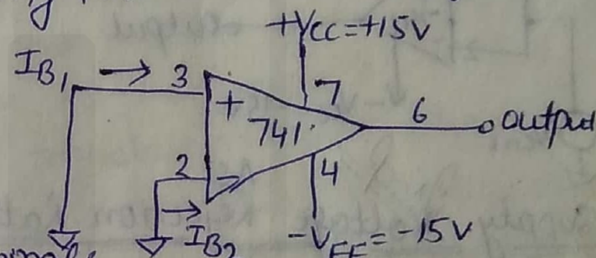


for 741C the max. value of V_{io} = 6mV dc.

Smaller the value of V_{io} better the input terminals are matched.

(3) DC **Input offset Current** (bias currents):- The algebraic difference between the currents into inverting and noninverting terminals. is refered as input offset current I_{io}.

$$\boxed{I_{io} = I_{B1} - I_{B2}}$$



smaller the I_{io} better the input terminals are matched. for 741C is 200nA max. 6uA

DC (2) **Input Bias Current**:- Input bias I_B is the average of currents that flow into inverting and noninverting input terminals of op amp.

$$I_B = \frac{I_{B1} + I_{B2}}{2}$$

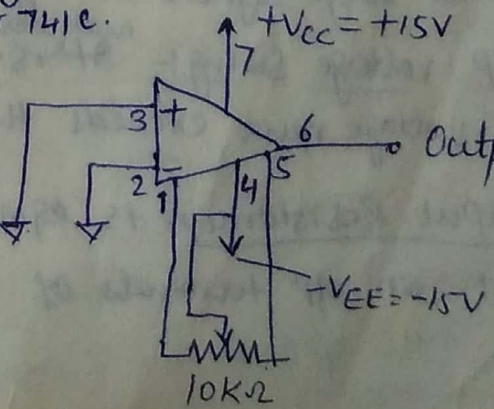
I_B = 500 nA. I_{B1} & I_{B2} are base c/n of first diff. amp. stage.

(4) DC char: **Thermal Drift**:- off c/n & offset volt. change with temp. eg. we have nulled at 25°C. Later at 40°C V_{io} is no longer nulled. It is called thermal drift.

Differential Input Resistance (R_i):- It is the equivalent resistance that can be measured at either inverting & noninverting input terminal with other terminal connected to ground. for 741C it is 2MΩ.

Input Capacitence (C_i):- It is the equivalent capacitence that can be measured at either inverting & noninverting terminal with the other terminal connected to ground. 1.4pF for 741C.

Offset Voltage Adjustment Range:- is the range through which input offset voltage can be adjusted by varying the 10kΩ potentiometer. output offset voltage can be reduced to zero volt. for 741C the offset volt. adjustment Range is ±15mV.



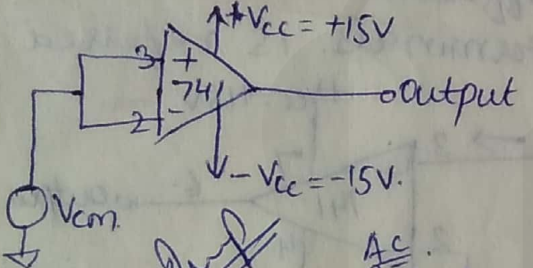
Common Mode Rejection Ratio :-

It is defined as the ratio of differential Voltage gain A_d to the common voltage gain A_{cm} i.e.

$$CMRR = \frac{A_d}{A_{cm}} \qquad A_{cm} = \frac{V_{ocm}}{V_{cm}}$$

V_{ocm} = O/P C.m. voltage
V_{cm} = I/P " "
A_{cm} = Common mode Voltage Gain

Generally A_{cm} is very small & CMRR is very large.
It is expressed in decibels (dB). For 741C CMRR is 90dB typically.
The higher the value of CMRR, the better is matching bet. two input terminals & smaller is the output common mode voltage.



Supply ~~Voltage~~ Rejection Ratio :-

It is the change in input offset voltage V_{io} caused by variation in supply voltage is called supply voltage rejection ratio. (SVRR). It is called power supply rejection ratio (PSRR) or power supply sensitivity (PSS).

$$SVRR = \frac{\Delta V_{io}}{\Delta V} \ \mu V/V \qquad SVRR = 20\log(\Delta V$$

Lower the value of SVRR in microvolts/volt, the better for opamp performance.

Large Signal Voltage Gain :-

opamp amplifies the difference bet. two i/p terminals.

$$A_o = \text{Voltage Gain} = \frac{\text{Output voltage}}{\text{differential i/p voltage}} = \frac{V_o}{V_{id}} = 2,00,000 \ (741C) \qquad V_o = \pm 10V$$

∵ O/P signal amp. is much larger than i/p signal, the voltage gain is called large signal voltage gain.

O/P voltage Swing :-

It is indicates the value of +ve & -ve sat. voltage of op amp. O/P voltage never exceeds these limits for given supply voltages +Vcc & -VEE. ±13V for $R_L \geq 2K\Omega$

Output Resistance :-

is equivalent resistance that can be measured between O/P terminals of op amp & the ground. It is 75Ω for 741C op amp.

Supply Current :- I_s is the current drawn by the op amp from the power supply. for 741C op-amp the supply current $I_s = 2.8\,mA$.

Power Consumption :- P_c is the amount of quiescent power ($V_{in} = 0V$) that must be consumed by op amp in order to operate properly. The amount of power consumed by 741C is 85mW.

AC

Transient Response :- The response of any practically useful N/w to a given I/P is composed of two parts : transient & steady state response. Transient response :- is that portion of complete response before O/P attains some fixed value. Once reached this fixed value remains at that level & is referred as steady state value. The response of N/w after it attains a fixed value is independent of time & is called steady state response. Transient response is time variant.

2 Rise time & % of overshoot are char. of transient response. The time required by O/P to go from 10% to 90% of its final value is called rise time. Overshoot is max. amount by which O/P deviates from steady state value. It is expressed as a %. Rise time $\propto \frac{1}{\text{unity Gain BW}}$. Smaller the rise time higher is BW

AC

3 Slew Rate :- It is the max. rate of change of O/P voltage per unit time & is expressed volts per microseconds. $V_{in} = V_p \sin\omega t$

$V_{out} = V_{pout} \sin\omega t$

$$SR = \frac{dV_o}{dt}\Big|_{max} \quad V/\mu s.$$

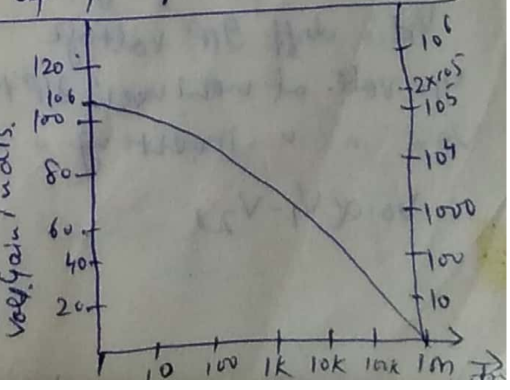
$$\left.\frac{dv_{out}}{dt}\right|_{max} = V_{pout}\,\omega\cos\omega t$$

$$= V_{pout}\,\omega = 2\pi f V_{pout} = SR.$$

Slew rate indicates how rapidly the output of op amp can change in response to change in the input frequency. Slew rate is one of the important factors in selecting the op amp for ac applications. 741C has low slew rate (0.5V/µs) which limit its use at high freq. applications i.e. oscillators, filters, comparators etc. LM318 has slew rate 70V/µs.

AC 1

Gain Bandwidth product :- GB product is the BW of op amp when voltage gain is 1. for an ideal op amp. open loop Gain is very large & does not vary with freq. Not These assumptions are not good for an actual op amp. The dc & very low freq. open loop gain is abt. 2×10^5 which is large. with increase in freq. the gain decreases prop. 741C has a critical freq. fc of 10Hz. Above fc gain decreases 20dB/decade untill it reaches 1 at freq. of 1MHz.



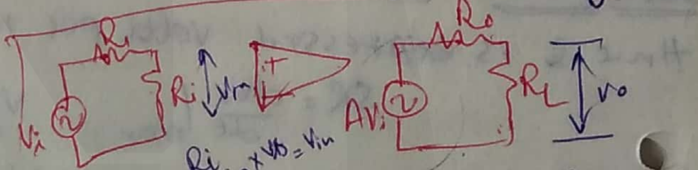
[Unity gain freq. f_{unity} is the freq. where voltage gain is unity. f_{unity} is 1MHz. 741C f_{unity} is 1MHz. So 741C can amplify the signal upto 1MHz. Beyond 1MHz voltage gain is less than unity & it is useless.]

$SR = \frac{dV_{out}}{dt}\Big|_{max}$. $V_{in} = V_{pin} \sin\omega t$ $V_{out} = V_{pout} \sin\omega t$

$\frac{dV_{out}}{dt} = V_{pout}\,\omega \cos\omega t$ $\frac{dV_{out}}{dt}\Big|_{max} = V_{pout}\,\omega = 2\pi f V_{pout}$ volt/sec

Ideal op-amp:-

1. Open loop gain is ∞ $V_{id} = 0$
2. Infinite i/p resistance means i/p current (current drawn from source) is zero. & so it does not load the source. so that almost any signal source can drive it. Ideal diode is voltage controlled device
3. Output resistance is zero so that o/p can drive an ∞ no. of other devices. Vout is independent of current drawn by the load.
4. Zero output voltage when i/p voltage is zero. (perfect balance).
5. Infinite BW so that any freq. signal from 0 to ∞ Hz can be amplified without attenuation (Dc and ac both signal can be amplified)
6. ~~Common~~ CMRR is ∞, so that amplifier is free from ~~common~~ thermal noise, pick up etc.
7. Slew rate is ∞ so that output voltage change occurs simultaneously with i/p voltage changes.
8. Output voltage is zero when i/p voltage is zero i.e. offset voltage is zero.
9. Drift of char. with temp. is nil.

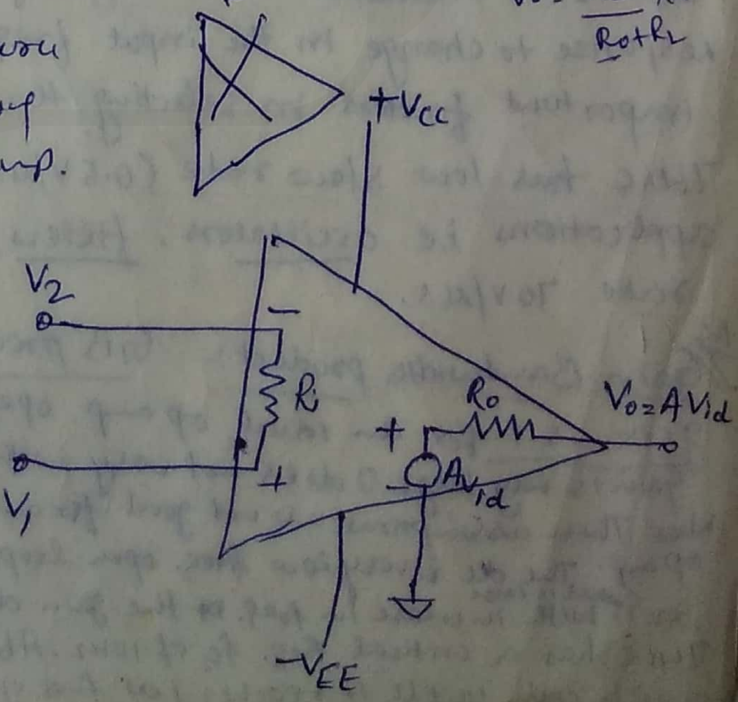


$\frac{R_i}{R_s+R_i} \times V_s = V_{in}$ $V_o = \frac{R_L}{R_o+R_L} \times AV_i$

(1) Equivalent ckt of an op amp:-

AV_{id} is ~~thevenin~~ an Thevenin equivalent voltage source

R_o is thevenin equivalent resistance looking back into the o/p terminal of an op-amp.



$V_o = AV_{id} = A(V_1 - V_2)$

A = Voltage gain

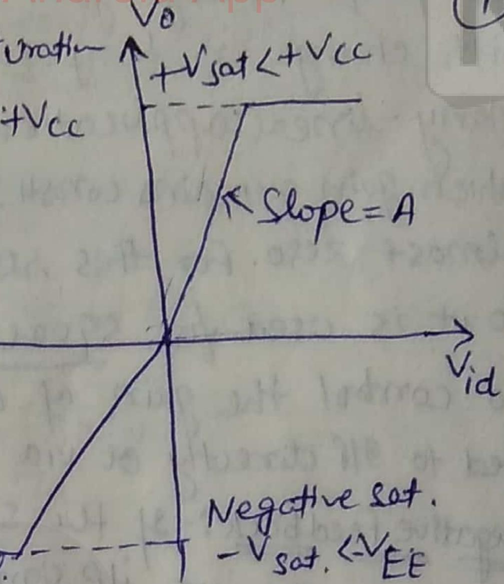
V_{id} = diff. i/p voltage

V_1 = volt. of noninverting i/p term w.r.t. ground

V_2 = " " inverting " " " "

$V_o \propto (V_1 - V_2)$

Ideal Voltage Transfer Curve :-



's ideal voltage transfer curve $+V_{sat} < +V_{cc}$

eal :: o/p offset voltage is zero. Gain const.

curve will be almost vertical because of very large value of A.

Open loop indicates there is no connection either direct or via another N/w exists bet. o/p & i/p terminal.

Open loop op amp Configurations :- means there is no f/b

In open loop it is high gain amplifier.

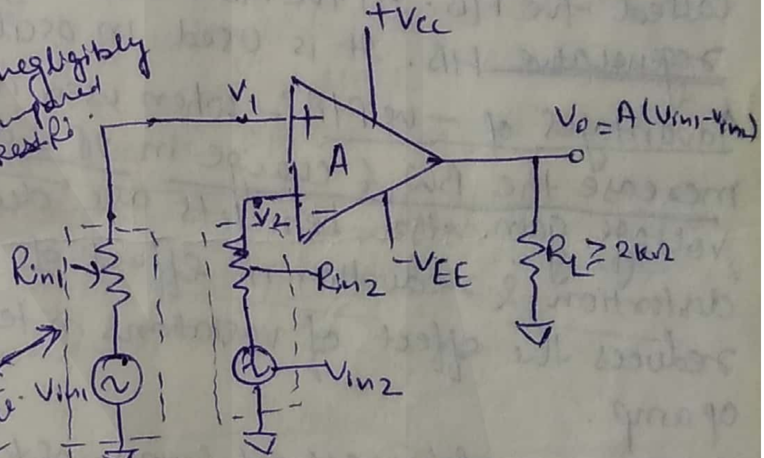
Three open loop config
- Differential amplifier
- Inverting "
- Non inverting amplifier.

Differential Amplifier :-

$V_1 = V_{in1}$ & $V_2 = V_{in2}$ R_{i1} & R_{in2} are negligibly small as compared to i/p Res. R_i

$V_0 = A(V_{in1} - V_{in2})$

$= A(V_1 - V_2)$



polarity of o/p voltage is dependent on polarity of I/P diff. voltage ($V_{in1} - V_{in2}$) Signal source.

Gain A is referred as open loop gain.

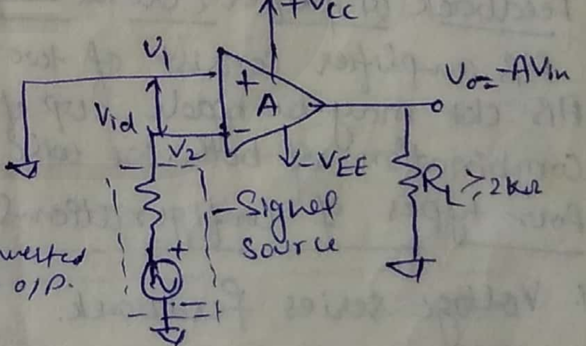
Inverting Amplifier :-

$V_1 = 0V$ & $V_2 = V_{in}$

$V_0 = -AV_{in}$

-ve sign indicates that o/p voltage is out of phase w.r.t. by 180° or is of opp. polarity.

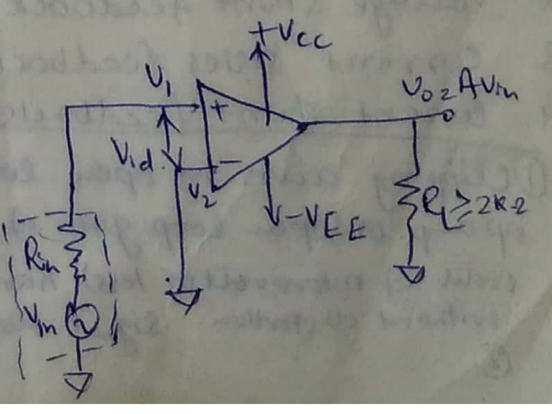
In inv. amp. i/p signal is amplified by gain A & is also inverted at o/p.



Non inverting Amplifier :-

$V_1 = V_{in}$ $V_2 = 0V$.

$V_0 = AV_{in}$



(1) In open loop config. any i/p signal that is only slightly greater than zero drives the o/p to sat. level. This is due to very high gain (A) of op amp. Thus, when operated open loop, the o/p of op amp is either -ve or +ve sat. or switches bet. +ve & -ve sat. levels. For this reason, open loop op-amp config. are not used in linear applications.

(2) Open loop voltage gain of op amp is not constant it varies with change in temp. & power supply. So it is unsuitable for many linear applications & (3) The bandwidth (band of freqs. for which gain remains constt.) of most open loop op amp is negligibly small almost zero. For this reason open loop op amp is ~~neg~~ impractical in ac applications So it is used for square wave applications.

To control the gain of op amp we use the feedback i.e. an o/p signal is fed to i/p directly or via another N/w.

Negative feedback :- If the signal fed back is of opposite polarity or out of phase by 180° wrt. i/p signal then it is called -ve feedback (degenerative feedback) ∵ when it is used it degenerates (reduces) the o/p voltage amplitude & in turn reduces the voltage gain.

+ve feedback :- If the signal fed back is in phase with i/p signal F/B is called +ve F/B. In +ve F/B the F/B signal aids the i/p signal. also called regenerative F/B. It is used in oscillator circuits.

Advantages of -ve F/B :- When used in amplifiers, -ve F/B stablizes the gain, increase the BW & change in i/p & o/p resistances. but it reduces the voltage gain. other benefits are decrease in nonlinear & harmonic distortion & reduction in effect of i/p offset voltage at the o/p. It also reduces the effect of variations in temp. & supply voltage on the o/p of op amp.

Feedback amplifier (Closed loop amplifier) :- Amp

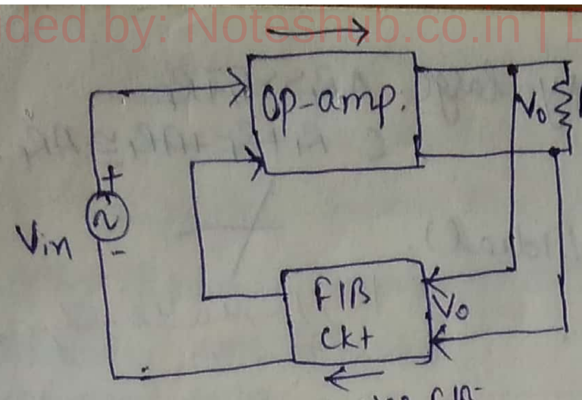
F/B amplifier consist of two parts: an opamp & a feedback circuit. F/B ckt may be made up of either passive or active components or combination of both. We will use only purely resistive F/B ckt.

Four types of configurations :-

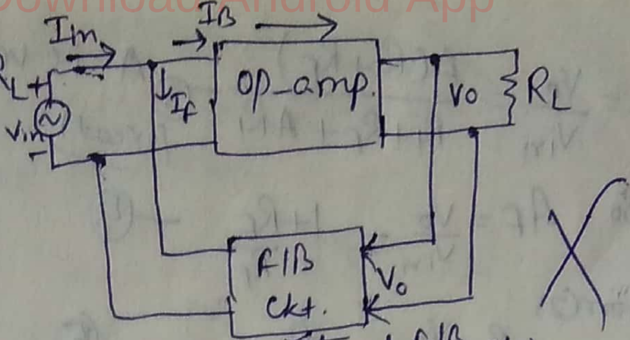
1. Voltage series feedback.
2. Voltage shunt feedback.
3. Current series feedback.
4. Current shunt feedback.

(1) Clipping occurs in open loop confi. when o/p exceeds the sat. level of op amp. or open loop gain of opamp is very high (only smaller signal (of order of microvolt or less) having very low freq. may be amplified accurately without distortion. Signal this small are very suscep

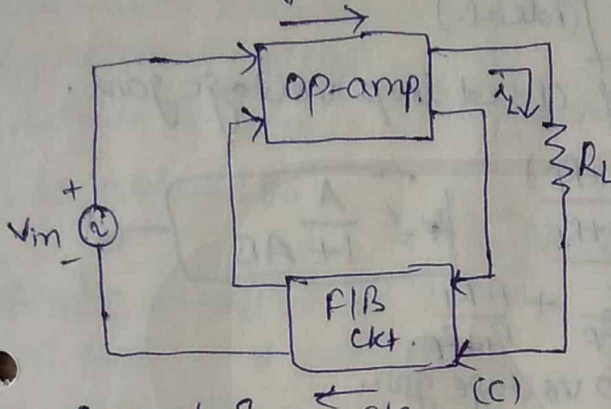
(2)



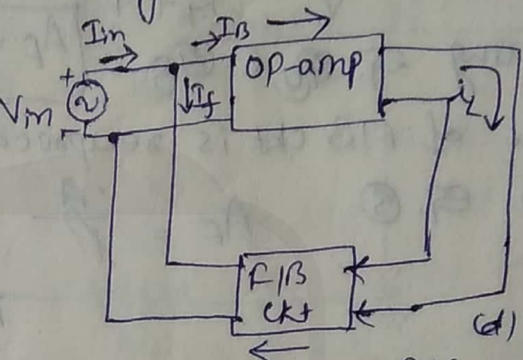
Voltage series F/B (a)



Voltage Shunt F/B. (b)



Current Series F/B (c)



(d) Current Shunt F/B.

In fig. (a) & (b) voltage across R_L is i/p voltage to F/B ckt. The F/B qty (volt. or c/u) is o/p of F/B ckt & is pro. to o/p voltage.

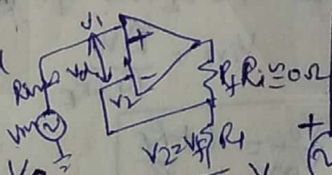
In fig. (c) & (d) the load c/u i_L flows into the F/B ckt. The o/p of F/B ckt (volt. or c/u) is pro. to load current i_L.

~~Voltage Series F/B Amplifier~~

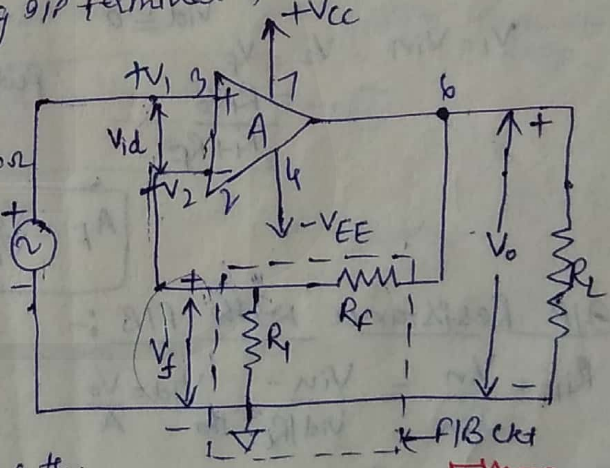
Non inverting Amplifier :- (Closed loop non inverting amplifier with F/B) :- It uses the F/B & i/p signal is applied to noninverting i/p terminal of op-amp.

open loop voltage gain $A = \frac{V_o}{V_{id}}$

closed " " " $A_F = \frac{V_o}{V_{in}}$



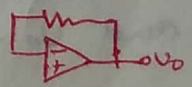
Gain of F/B circuit $B = \frac{V_f}{V_o}$.



Apply KVL for i/p loop is

$$V_{id} = V_{in} - V_f \quad — ①$$

In above eqn. F/B voltage always oppose the i/p volt. hence F/B is -ve.



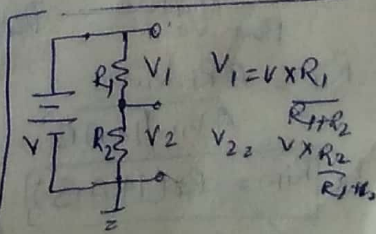
We know $V_o = A(V_1 - V_2)$ $\quad V_1 = V_{in} \quad V_2 = V_f = \frac{R_1 V_o}{R_1 + R_F} \quad — ②$

$$V_o = A\left(V_{in} - \frac{R_1 V_o}{R_1 + R_F}\right)$$

$$V_o = A\left(\frac{V_{in}(R_1 + R_F) - R_1 V_o}{R_1 + R_F}\right)$$

$$V_o\left(\cancel{A}1 + \frac{AR_1}{R_1 + R_F}\right) = AV_{in}$$

$$V_o\left(\frac{R_1 + R_F + AR_1}{R_1 + R_F}\right) = AV_{in}$$



$V_1 = V \times \frac{R_1}{R_1 + R_2}$

$V_2 = V \times \frac{R_2}{R_1 + R_2}$

$A_f = \frac{V_o}{V_{in}} = \frac{A(R_1+R_f)}{R_1+R_f+AR_1}$ —③ (exact)

A is very large. $AR_1 >> R_1+R_f$ & $R_1+R_f+AR_1 \cong AR_1$

So $A_f = \frac{V_o}{V_{in}} = 1+\frac{R_f}{R_1}$ —④ (ideal).

~~17, 18, 40, 46, 51~~, 54

from (2)

$B = \frac{V_f}{V_o} = \frac{R_1}{R_1+R_f}$ —⑤

Comparing eq. ④ & ⑤ $\boxed{A_f = \frac{1}{B}}$ (ideal.)

So Gain of F/B ckt is reciprocal of closed loop voltage gain.

from eq. ③ $A_f = \dfrac{\frac{A(R_1+R_f)}{R_1+R_f}}{\frac{R_1+R_f}{R_1+R_f}+\frac{AR_1}{R_1+R_f}}$ $\boxed{A_f = \frac{A}{1+AB}}$ —⑥

$\boxed{A_f = \frac{A}{1+AB}}$ —⑥

A_f = Closed loop voltage gain

A = open " " "

B = Gain of F/B ckt.

AB = loop gain.

Difference input voltage ideally zero:-

$V_{id} = \frac{V_o}{A}$ Since A is very large (ideally ∞)

$V_{id} \cong 0$ i.e. $V_1 \cong V_2$ (ideal) —(a)

$V_1 = V_{in}$ $V_2 = V_f = \frac{R_1 V_o}{R_1+R_f}$

Put V_1 & V_2 in eq. (a)

$V_{in} = \frac{R_1 V_o}{R_1+R_f}$

$\boxed{A_f = \frac{V_o}{V_{in}} = 1+\frac{R_f}{R_1}}$

I/P Resistance with F/B:-

$R_{if} = \frac{V_{in}}{I_{in}} = \frac{V_{in}}{V_{id}/R_i}$ —(b) $V_{id} = \frac{V_o}{A}$

and $V_o = \frac{A}{1+AB} V_{in}$

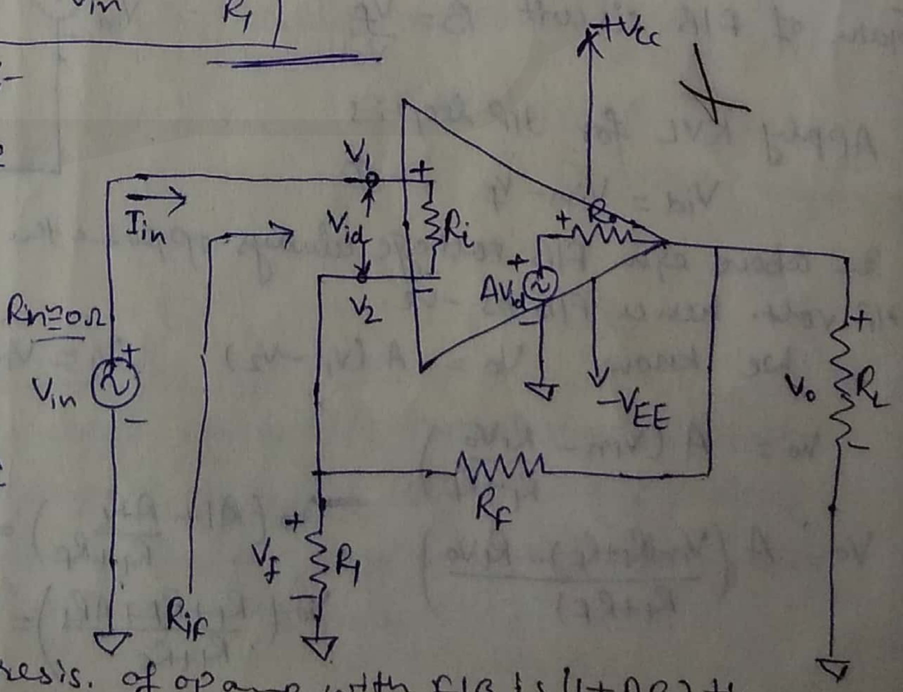
Put V_{id} in eq.(b)

$R_{if} = \frac{R_i V_{in}}{V_o/A}$ $R_{if} = \frac{AR_iV_{in}}{V_o}$

Put V_o.

$R_{if} = \frac{AR_iV_{in}}{AV_{in}/1+AB}$

So $\boxed{R_{if} = R_i(1+AB)}$

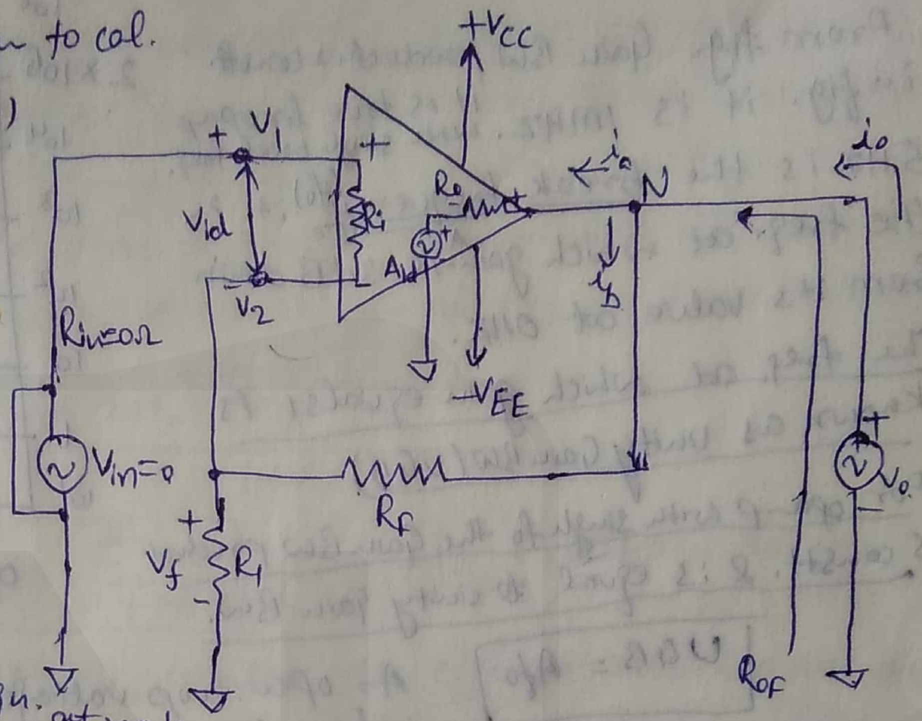


So I/P resis. of op amp with F/B is (1+AB) times that without F/B.

Output Resistance with feedback :-

It is the resistance determined look back into the F/B amplifier from the o/p terminal.

We apply Thevenin Theorem to cal. o/p resistance with F/B. (R_{oF}) for dependent source. (independent source) Reduce V_{in} to zero & apply external voltage V_o & then cal. i_o.



R_{oF} is defined as

$$R_{oF} = \frac{V_o}{i_o} \quad \text{—(1)}$$

Write Kirchoff's c/n equ. at node N.

$$i_o = i_a + i_b$$

Since $(R_F + R_1) \| R_i >> R_o$ & $i_a >> i_b$

$$\therefore \; i_o \simeq i_a$$

Current i_o can be found by writting Kirchoff's voltage eqn. from o/p loop.

$$V_o - R_o i_o - AV_{id} = 0 \qquad \boxed{i_o = \frac{V_o - AV_{id}}{R_o}} \quad \text{—(2)}$$

$$V_{id} = V_1 - V_2$$

$$V_{id} = 0 - \frac{R_1 V_o}{R_1 + R_F} = -BV_o.$$

Put V_{id} in eqn. (2)

$$\therefore \; i_o = \frac{V_o + ABV_o}{R_o}$$

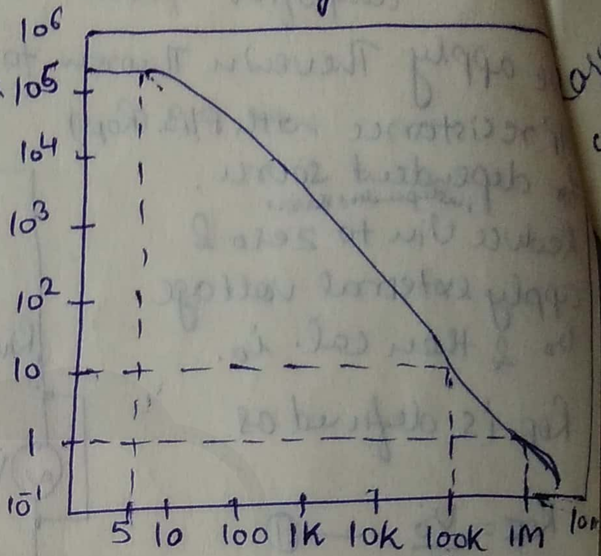
Put value in eq. (1)

$$R_{oF} = \frac{V_o}{(V_o + ABV_o)/R_o} = \qquad R_{oF} = \frac{R_o}{1+AB}$$

So o/p resistance of voltage series F/B amp. is $\frac{1}{1+AB}$ times the o/p resistance R_o of the op amp. i.e. o/p resistance of op amp with F/B is much smaller than o/p resistance without F/B.

Bandwidth with feedback :-

BW is defined as the band (range) of freqs. for which Gain remains constt.



open loop gain vs freq curve for 741C.

From fig. Gain BW product is constt. in fig. it is 1MHZ. it is true for opamp with single break freq. 5Hz is the break frequency (f_0), it is the freq. at which gain A is 3dB down from its value at 0HZ.

The freq. at which gain equals 1 is known as unity Gain BW (UGB)

For opamp with single f_0 the Gain BW product is constt. & is equal to unity gain BW.

$$\boxed{UGB = Af_0}$$

A = open loop voltage gain

f_0 = break freq. of an op amp.

Or only for a single break frequency op amp.

$$\boxed{UGB = (A_F)(f_F)}$$

A_F = Closed loop voltage gain

f_F = BW with F/B.

$$Af_0 = A_F f_F$$

$$f_F = \frac{Af_0}{A_F}$$

For non inverting amplifier with F/B. $A_F = \dfrac{A}{1+AB}$

Substitute A_F in above eqn.

$$f_F = \frac{Af_0}{A/1+AB} = \boxed{f_F = f_0(1+AB)}$$

eqn. show that BW of noninverting amplifier with feedback f_F is equal to $(1+AB)$ times to its BW without F/B (f_0).

Total Output offset voltage with F/B :-

In an op-amp when i/p is zero, o/p is also expected to zero. ∵ of effect of i/p offset voltage & c/n, the o/p is significantly larger ∵ of very high open loop gain. We call this enhanced o/p voltage is called total o/p offset voltage V_{OOT}. In open loop op-amp total o/p offset voltage is either equal to +ve or -ve sat. voltage.

Due to F/B gain of non-inver. amp. changes from A to $A/(1+AB)$. So total o/p offset voltage with F/B must also be $1/(1+AB)$ times the volt. without F/B.

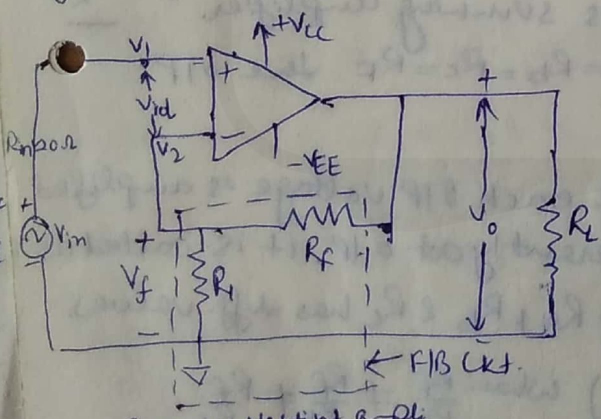
$$\text{Total o/p offset voltage with F/B} = \frac{\text{Total o/p offset voltage without F/B}}{1+AB}$$

$$\boxed{V_{OOT} = \frac{\pm V_{sat}}{1+AB}}$$

$1/1+AB$ is always less than 1.

Higher the value of $(1+AB)$, smaller the effect of noise & of variations in supply voltages & change in temp. on o/p voltage of a noninverting amplifier.

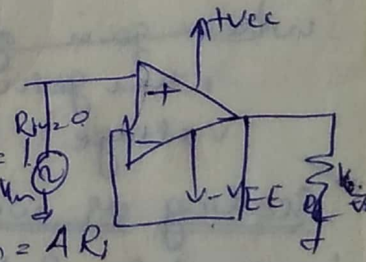
Voltage follower:- When the noninverting amplifier is configured for unity gain is called a voltage follower ∵ o/p voltage is equal to & in phase with i/p. In voltage follower the o/p follows the i/p. It is similar to discrete emitter follower. The voltage follower is preferred ∵ it has much higher i/p resi. & o/p ~~amplifier~~ amplitude is exactly to i/p.



non inverting ampli.

open R_1 & short R_F

$B = 1$



Non-inverting amp.	Voltage follower
$A_f = \dfrac{A}{1+AB}$	$A_F = 1$
$R_{iF} = R_i(1+AB)$	$R_{iF} = AR_i$
$R_{oF} = \dfrac{R_o}{1+AB}$	$R_{oF} = \dfrac{R_o}{A}$
$f_F = (1+AB)f_o$	$f_F = Af_o$
$V_{OOT} = \dfrac{\pm V_{sat}}{1+AB}$	$V_{OOT} = \dfrac{\pm V_{sat}}{A}$

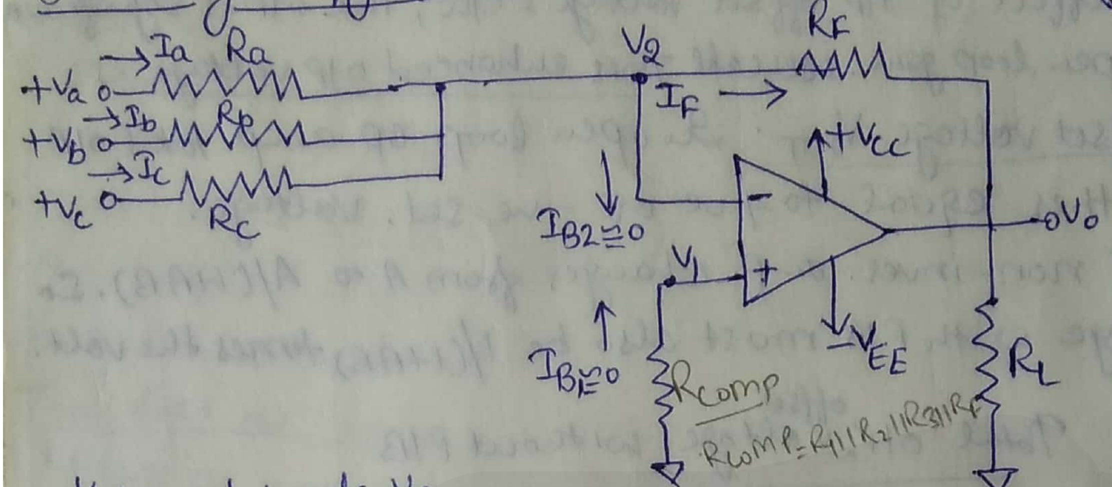
$V_o = V_{in}$

Since $(1+A) \simeq A$

Volt. foll. is also called noninverting buffer ∵ when placed between two N/ws it removes the loading on first N/w.

Summing, Scaling & Averaging Amplifier :-

Inverting Configuration :-



KCL at node V_2

$$I_a + I_b + I_c = I_B + I_F, \quad \therefore \frac{V_a - V_2}{R_a} + \frac{V_b - V_2}{R_b} + \frac{V_c - V_2}{R_c} = \frac{V_2 - V_o}{R_F}$$

since R_i & A are ideally ∞, $I_B = 0A$, & $V_1 = V_2 \cong 0V$

$$\frac{V_a}{R_a} + \frac{V_b}{R_b} + \frac{V_c}{R_c} = \frac{-V_o}{R_F}$$

$$V_o = -\left(\frac{R_F V_a}{R_a} + \frac{R_F}{R_b} V_b + \frac{R_F}{R_c} V_c\right)$$

[gain Inverting : $A_d = -\frac{R_f}{R_1}$]

Summing amplifier :- If $R_a = R_b = R_c = R$

then $V_o = -(V_a + V_b + V_c)\frac{R_F}{R}$

So o/p voltage is -ve sum of all i/ps times the gain of ckt $\frac{R_F}{R}$. hence it is summing amplifier.

~~when gain of ckt is 1.~~ when gain of ckt is 1. i.e. $R_a = R_b = R_c = R_F$ the o/p voltage $V_o = -(V_a + V_b + V_c)$.

Scaling or weighted Amplifier :- If each i/p voltage is amplified by a diff. factor or weighted differently at o/p, it is called as Scaling or weighted amplifier. If R_a, R_b & R_c has diff. values.

$$V_o = -\left(\frac{R_F}{R_a} V_a + \frac{R_F}{R_b} V_b + \frac{R_F}{R_c} V_c\right) \text{ when } \frac{R_f}{R_a} \neq \frac{R_f}{R_b} \neq \frac{R_f}{R_c}$$

Average circuit :- in which o/p volt. is equal to avg. of all i/p voltages. So $R_a = R_b = R_c = R$ or gain by which each i/p is amplified must be equal to 1 over no. of i/ps i.e.

$\boxed{\frac{R_F}{R} = \frac{1}{n}}$, where n = no. of i/ps. $\quad \frac{R_f}{R} = \frac{1}{3}$

$V_o = -\left(\frac{V_a + V_b + V_c}{3}\right)$

In ckt we want $\frac{R_f}{R} = \frac{1}{3}$ So

Imp

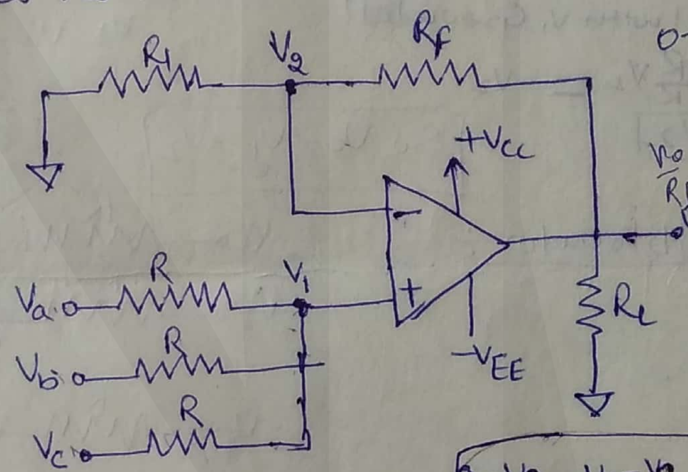
So $V_o = -\left(\frac{V_a + V_b + V_c}{3}\right)$

Use: These ckts are used in analog computers & audio mixers in which a no. of I/Ps are added up (mixed) up to produce a desired O/P.

In practical ckt I/P bias c/n compensating resistor R_{comp} should be provided. To find R_{comp} make all I/Ps $V_1 = V_2 = V_3 = 0$ so that effective I/P resis. $R_i = R_1 \| R_2 \| R_3$ ∴ $R_{comp} = R_i \| R_f = R_1 \| R_2 \| R_3 \| R_f$.

Non Inverting Summing Amplifier:-

A summer that gives non inverted sum is non inverting summing amplifier



$\frac{0-V_2}{R_1} + \frac{V_2 - V_o}{R_f} = 0$

$\frac{V_o}{R_f} = \frac{V_2}{R_1} + \frac{V_2}{R_f}$

$-R_f V_2 + R_1 V_o - V_2 R_1 = 0$

$R_1 V_o = V_2 R_1 + R_f V_2$

$V_o = V_2\left(1 + \frac{R_f}{R_1}\right)$

$$\frac{V_a - V_1}{R} + \frac{V_b - V_1}{R} + \frac{V_c - V_1}{R} = 0$$

So $V_1 = \frac{V_a + V_b + V_c}{3}$

$\boxed{\frac{0-V_2}{R_1} + \frac{V_o - V_2}{R_f} = 0}$ $V_o = V_2\left(1 + \frac{R_f}{R_1}\right)$

Hence O/P voltage is

$$\boxed{V_o = \left(1 + \frac{R_f}{R_1}\right) V_1}$$

$V_o = V_2\left(\frac{R_f}{R_1} + 1\right)$

$$\boxed{V_o = \left(1 + \frac{R_f}{R_1}\right)\left(\frac{V_a + V_b + V_c}{3}\right)}$$

Avg: Amplifier:- O/P voltage is equal to avg. of all I/P time gain of ckt $\left(1 + \frac{R_f}{R_1}\right)$

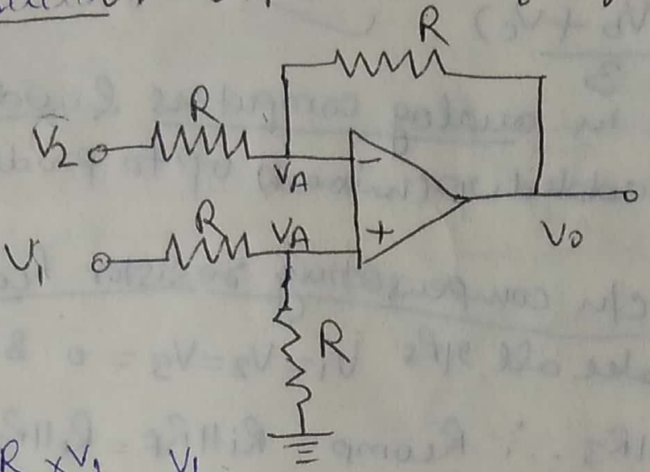
If the gain is 1, O/P voltage is equal to avg. of all I/P voltage.

Summing amplifier:- if $\left(1 + \frac{R_f}{R_1}\right)$ is equal to no. of I/Ps the O/P voltage becomes equal to sum of all I/P voltage. i.e. if $\left(1 + \frac{R_f}{R_1}\right) = 3$

So $\boxed{V_o = V_a + V_b + V_c}$

~~Voltage Shunt f~~ (Differential amplifier)

Subtractor:- & All R are of equal value



Apply Nodal

$$\frac{V_1 - V_A}{R} + \frac{0 - V_A}{R} = 0$$

$$\boxed{V_A = \frac{V_1}{2}}$$

$V_A = \frac{R}{R+R} \times V_1 \Rightarrow \frac{V_1}{2}$

V_1 alone make $V_2 = 0$

$V_{o1} = (1 + \frac{R}{R})\frac{V_1}{2} = V_1$

V_2 alone (with V_1 Grounded)

$V_{o2} = -\frac{R}{R} V_2 = -V_2$

$\boxed{V_o = V_1 - V_2}$

$$\cancel{\frac{V_1 - V_2}{R}} \quad \frac{V_2 - V_A}{R} + \frac{V_o - V_A}{R} = 0$$

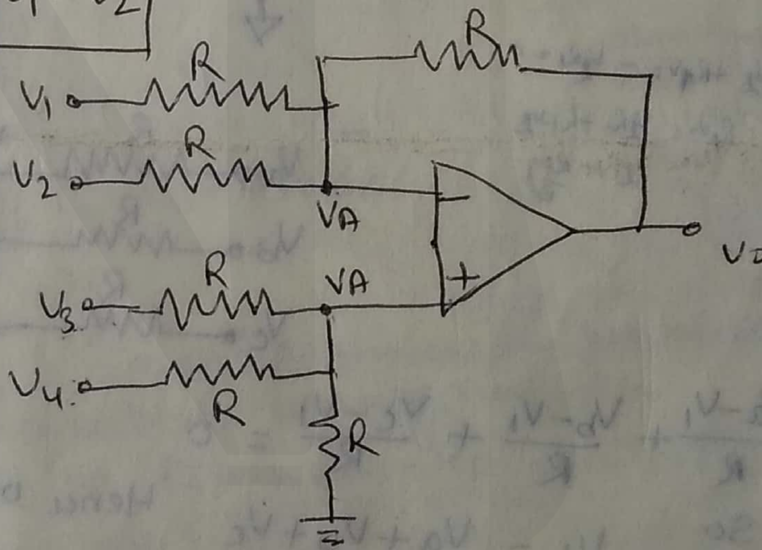
$V_2 + V_o - 2V_A = 0$

$V_o = 2V_A + V_2$ put V_A

$= 2 \times \frac{V_1}{2} - V_2$

$\boxed{So\ V_o = V_1 - V_2}$

Adder Subtractor:-



$$\frac{V_1 - V_A}{R} + \frac{V_2 - V_A}{R} + \frac{V_o - V_A}{R} = 0$$

$V_1 + V_2 + V_o - 3V_A = 0$

$\boxed{V_1 + V_2 + V_o = 3V_A}$ — (1)

$$\frac{V_3 - V_A}{R} + \frac{V_4 - V_A}{R} + \frac{0 - V_A}{R} = 0$$

$\boxed{V_3 + V_4 = 3V_A}$ — (2)

from (1) & (2)

$V_1 + V_2 + V_o = V_3 + V_4$

$$\boxed{V_o = (V_3 + V_4) - (V_1 + V_2)}$$